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Ex.No:1: Simulation of FDMA in MATLAB Date

Aim: To simulate a Frequency Division Multiple Access (FDMA) system using MATLAB

and analyze its communication performance in terms of channel bandwidth, data rate, and

signal-to-noise ratio (SNR).

Objective 1:

To simulate an FDMA system by allocating individual frequency channels to multiple users

and determine the channel bandwidth assigned to each user.

Output Parameter: Channel Bandwidth (kHz)

Objective 2:

To calculate the data rate achieved by each user in the FDMA system based on the allocated

channel bandwidth.

Output Parameter: Data Rate (kbps)

Objective 3:

To evaluate the communication quality of the FDMA system by calculating the Signal-to-

Noise Ratio (SNR) for different users.

Output Parameter: Signal-to-Noise Ratio (SNR) (dB)

Procedure

1. Open MATLAB and create a new script file.

2. Define the total available bandwidth, number of FDMA users, carrier frequencies, signal

power, and noise power.

3. Allocate a unique frequency channel to each user and calculate the channel bandwidth.

4. Compute the data rate for each user based on the allocated bandwidth and modulation

scheme.

5. Calculate the Signal-to-Noise Ratio (SNR) using the signal and noise power values.

6. Plot the transmitted signals of all FDMA users using different carrier frequencies.

7. Display the calculated channel bandwidth, data rate, and SNR in the MATLAB

Command Window.

8. Analyze the results to verify independent channel allocation and evaluate the overall

performance of the FDMA system.

Program (Matlab Code)

```
clc; clear; close all;

fs=1000; t=0:1/fs:1;

TBW=600; Users=3;


CBW=TBW/Users;

fc=[100 300 500];

m=sin(2*pi*10*t);

s1=m.*cos(2*pi*fc(1)*t);

s2=m.*cos(2*pi*fc(2)*t);

s3=m.*cos(2*pi*fc(3)*t);

subplot(2,1,1)

plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on

title('FDMA Channel Allocation'),xlabel('Time (s)'),ylabel('Amplitude')

subplot(2,1,2)

bar(1:Users,CBW*ones(1,Users)),grid on

xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')
```

Output:

Objective 1:

Output Graph 1: FDMA Channel Allocation

• Displays three users transmitting on different carrier frequencies (100 kHz, 300 kHz, and 500 kHz).

• Demonstrates that each user occupies a separate frequency channel.

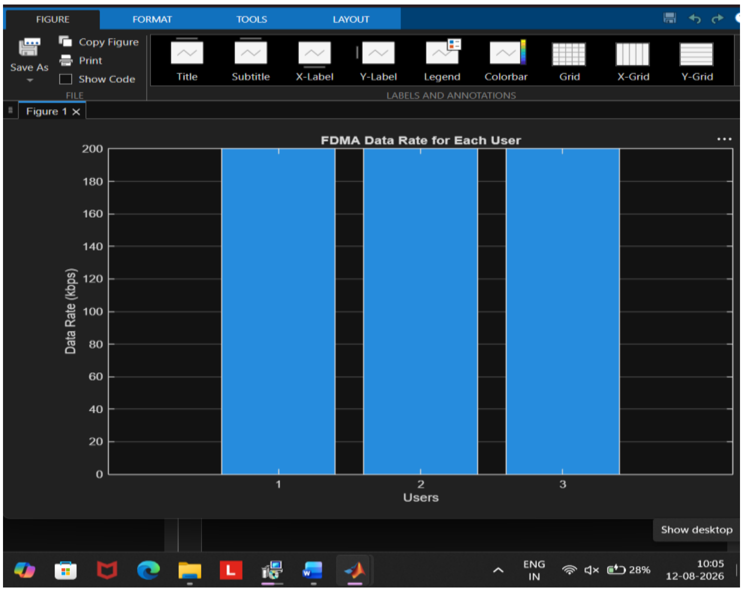
Output Graph 2: Channel Bandwidth Allocation

• Shows a bar graph of the bandwidth allocated to each user.

• Each user is assigned 200 kHz, confirming equal FDMA channel allocation.

Objective 2: Matlab Code and Ootput

```
clc; clear; close all;

BW = 200; % Channel Bandwidth (kHz)

Users = 3;

M = 2; % BPSK Modulation

DataRate = BW*log2(M); % Data Rate (kbps)

Rate = DataRate*ones(1,Users);

disp(' User Data Rate (kbps)')

disp([(1:Users)' Rate'])

figure

bar(1:Users,Rate)

xlabel('Users')

ylabel('Data Rate (kbps)')

title('FDMA Data Rate for Each User')

grid on
```

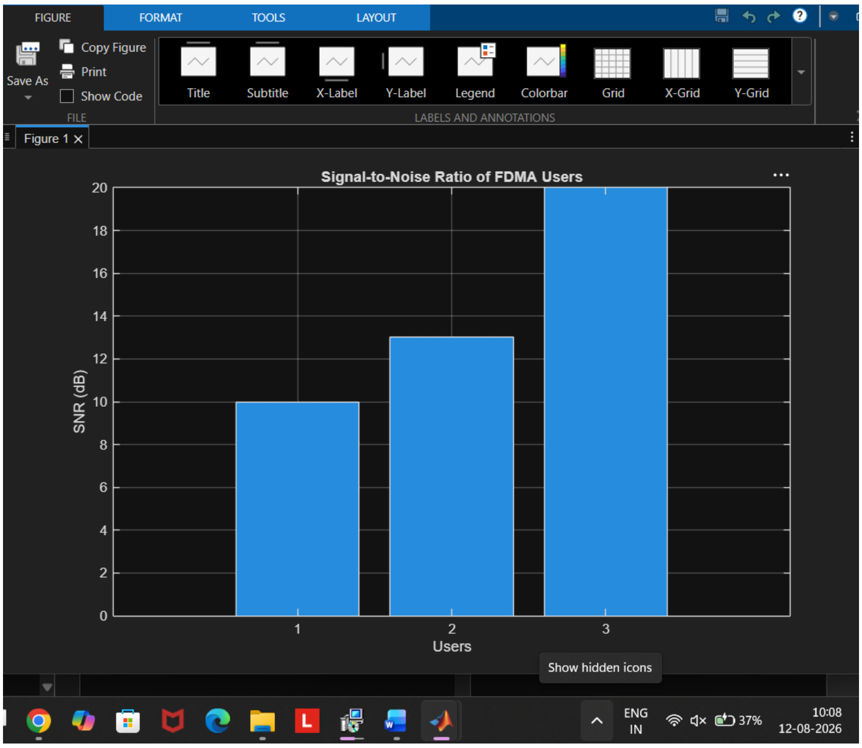


The graph shows that all FDMA users achieve an equal data rate of 200 kbps due to equal channel

bandwidth allocation. This confirms fair and interference-free data transmission among users.

Objective 3: Matlab Code and Ootput

```
clc; clear; close all;

Ps = 1; % Signal Power (W)

Pn = [0.1 0.05 0.01]; % Noise Power (W)

Users = 1:3;

SNR = 10*log10(Ps./Pn); % SNR in dB

disp('User Noise Power(W) SNR(dB)')

disp([Users' Pn' SNR'])

figure

bar(Users,SNR)

xlabel('Users')

ylabel('SNR (dB)')

title('Signal-to-Noise Ratio of FDMA Users')

grid on
```



The graph shows that the SNR increases as the noise power decreases for different FDMAusers. A higher SNR indicates better communication quality and more reliable signal

transmission.

Result:

1. The FDMA system was successfully simulated by allocating separate frequency channels

to three users, and the channel bandwidth assigned to each user was determined.

2. The data rate achieved by each user was successfully calculated based on the allocated

channel bandwidth, demonstrating equal transmission capacity for all users.

3. The Signal-to-Noise Ratio (SNR) for different users was successfully evaluated,

showing that higher SNR values provide better communication quality and reliable data

transmission.